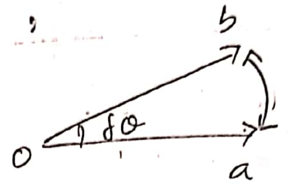
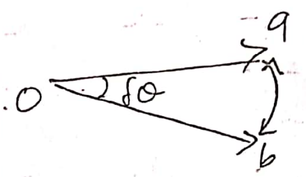
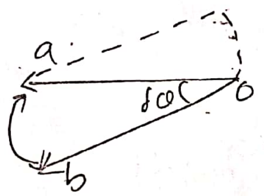
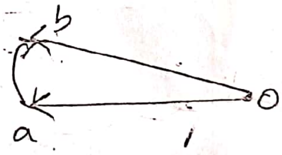
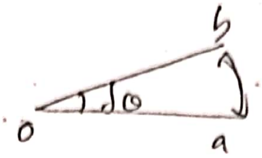
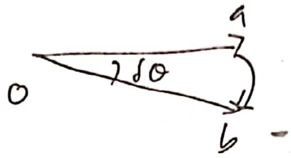
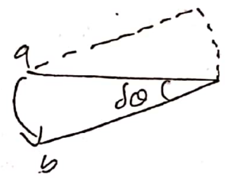
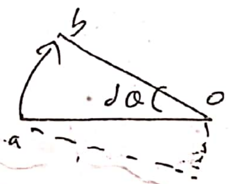
Gyroscopic Effect chart for Aeroplanes ①

View point	Direction of rotation of propeller	Turn	Effect
Rear - (or) Front -	clockwise (or) Anti clockwise	left	Nose raised Tail depressed
Rear - (or) Front -	clockwise (or) Anti clockwise	right	Nose depressed Tail raised
Rear - (or) Front -	Anti clockwise (or) clockwise	left	Nose depressed Tail raised
Rear - (or) Front -	Anti clockwise (or) clockwise	right	Nose raised Tail depressed

Gyroscopic effect chart for Naval ships during steering

View point	Direction of rotation of propeller	Steering	Vector diagram	Effect
Stern - Bow -	clockwise Anti clockwise	left		Bow raised Stern depressed
Stern - Bow -	clockwise Anti clockwise	right		Bow depressed Stern raised
Stern - Bow -	Anti clockwise Clockwise	left		Bow depressed Stern raised
Stern - Bow -	Anti clockwise clockwise	right		Bow raised Stern depressed

Gyroscopic effect chart for Naval ships during pitching

View point	Direction of rotation of propeller	Pitching	Vector diagram	Effect
Stern – Bow –	clockwise Anti clockwise	up wards		Ship turn towards Sternboard side Bow towards Starboard.
Stern – Bow –	clockwise Anti clockwise	Down wards		Ship turns towards portside
Stern – Bow –	Anticlock wise clockwise	up wards		Ship turns towards portside
Stern – Bow –	Anti clockwise clockwise	Down wards		Ship turns towards Starboard side

Gyroscopic of 4 wheel

1) Reaction due to the wt of the vehicle $= \frac{mg}{4} = \frac{W}{4}$

2) Reaction due to Gyroscopic couple

a) gyroscopic couple due to 4 wheels

$$C_W = 4 \times I_W \omega_W \omega_P$$

$$\omega_P = \frac{V}{R} \quad \omega_W = \frac{V}{r_W}$$

R = radius of curvature

b) couple due to rotating parts of engine

$$C_E = I_E \omega_E \omega_P$$

$$\omega_E = G \times \omega_W$$

Net gyroscopic couple $C = C_W \pm C_E$

+ wheel & rotating part rotating the same direction

− opposite

Reaction on outer wheel $P = \frac{C}{x}$ (upward)

inner wheel $P = \frac{C}{x}$ (downward)

$x \rightarrow$ width of track

3) Reaction due to centrifugal couple

$$Q = \frac{mv^2 h}{Rx}$$

4). total reaction..

$$P_0 = \frac{W}{4} + \frac{P}{2} + \frac{Q}{2} \quad \text{(outer)}$$

$$P_1 = \frac{W}{4} - \frac{P}{2} - \frac{Q}{2} \quad \text{(inner)}$$

(Q) A uniform disc of dia 30 cm & weighing 5N is mounted on one end of an arm of length 60 cm. The other end of the arm is free to rotate in a universal bearing. If the disc rotates about the arm with a speed of 300 rpm clockwise, looking from the front, with what speed will it precess about the vertical axis?

Ans

$d = 30\,cm, \quad r = 15\,cm$

$W = mg = 5N, \quad \therefore m = \frac{5}{9.81} = 0.509\,kg$

$l = 0.60\,m \quad N = 300\,rpm \quad \omega = \frac{2\pi N}{60} = 31.41\,r/s$

$I = \frac{mr^2}{2} = \frac{0.509 \times (0.15)^2}{2} = 5.726 \times 10^{-3}\,kgm^2$

$C = mgl$

$= 5 \times 0.6 = 3\,Nm$

$C = I\omega\omega_p$

$3 = 5.726 \times 10^{-3} \times 31.41 \times \omega_p$

$\omega_p = 16.68\,r/s$